



KCED
Learning Augmented

24ADI003-MACHINE LEARNING

My Ground rules are still the same!

- 1. Presence is Prerequisite: Attendance is Mandatory.**
- 2. Deadlines are Commitments: Respect the Schedule.**
- 3. Assignments are Training, Not Optional.**
- 4. Your Work Must Be Your Own: Uphold Academic Integrity.**
- 5. Assessments are Your Proof of Mastery.**
- 6. This is a Partnership: My Commitment to You.**

Course Objectives:

The purpose of taking this course is to:

1. Introduce the fundamental concepts of machine learning, its life cycle and ethical considerations.
2. Explore various supervised and unsupervised learning techniques and optimization strategies.
3. Examine recommendation systems, and its evaluation techniques.

Course Outcomes

- After successful completion of this course, the students shall be able to
- **CO1** Apply appropriate **data pre-processing techniques** to build machine learning models with ethical considerations. (Ap)
- **CO2** Analyze and optimize **regression models using estimation techniques, regularization, and gradient-based methods** through error analysis. (An)
- **CO3** Build and evaluate the effectiveness of **different classification models and ensemble techniques**. (E)
- **CO4** Analyze complex datasets using advanced **clustering, associative rule mining and dimensionality reduction algorithms** to uncover meaningful patterns and groupings. (An)
- **CO5** Develop **recommendation systems** to personalize user needs. (Ap)

Course Content

• INTRODUCTION

Introduction to Machine Learning-Types of machine learning: Supervised, Unsupervised, Semi supervised and Reinforcement Learning-Applications of machine learning in various fields- Ethics in machine learning- Fairness, accountability and interpretability -Machine learning workflow- Data Preprocessing-Feature engineering- Correlation analysis- Model training and evaluation- Model monitoring and maintenance.

• REGRESSION MODELS

Linear regression- Simple Regression-Least Square Estimator-Maximum Likelihood Estimator- Multiple Regression -Polynomial Regression-Performance Metrics-Bias

• CLASSIFICATION MODELS

Logistic Regression-Naive Bayes Classifiers-Decision Tree-K-Nearest Neighbors-Support Vector Machine - Evaluation metrics - AUC ROC- Class Imbalance -SMOTE - Cross-Validation Techniques-Ensemble Learning- Bagging- Random Forests-Boosting - AdaBoost -Gradient Boosting

• UNSUPERVISED LEARNING

Clustering- K-means Clustering- Gaussian Mixture Models -Hierarchical Clustering- Density-Based Clustering (DBSCAN)- Mean-Shift Clustering- Spectral Clustering- Association Rule Learning- Apriori Algorithm- FP-Growth Algorithm- Dimensionality Reduction- Principal Component Analysis (PCA)- Linear Discriminant Analysis (LDA)

• RECOMMENDATION SYSTEMS

Introduction to Recommendation Systems-Types- Challenges- Collaborative Filtering Techniques- User-Based- Collaborative Filtering- Item-Based- Collaborative Filtering- Matrix Factorization Techniques- Content-Based Recommendation- Hybrid Recommendation Systems- Evaluation of Recommendation Systems

Unit I – Introduction

Introduction to Machine Learning-Types of machine learning: Supervised, Unsupervised, Semi supervised and Reinforcement Learning-Applications of machine learning in various fields- Ethics in machine learning- Fairness, accountability and interpretability - Machine learning workflow- Data Preprocessing-Feature engineering- Correlation analysis- Model training and evaluation- Model monitoring and maintenance.

ML Helps Oil Companies Strike More Efficiently

- In late **2025**, *BP* — one of the world's largest energy companies — publicly credited **advanced artificial intelligence (AI) and ML systems** for transforming how it discovers oil and gas reserves, dramatically improving exploration success and operational efficiency.
- Instead of relying solely on traditional physics-based and geological analysis (which can take months to interpret), BP brought in ML technologies that can:
 - Process **millions of sensor readings** from subsurface and seismic data
 - Build a **digital twin** of operations
 - Predict with high precision where oil reserves are likely to be found
 - Run thousands of simulation models *much faster* than before
 - Reduce time from data interpretation to actionable insights by an order of magnitude
- **In essence**, BP used ML to *do in days what once took months or years*, with higher accuracy and lower costs.

MACHINE LEARNING: THE NEW 'OIL'



- **FASTER DECISION-MAKING**
- **HIGHER ACCURACY**
- **PREDICTED OIL RESERVES**
- **REDUCED EXPLORATION COST**
- **IMPROVED SUCCESS RATE**

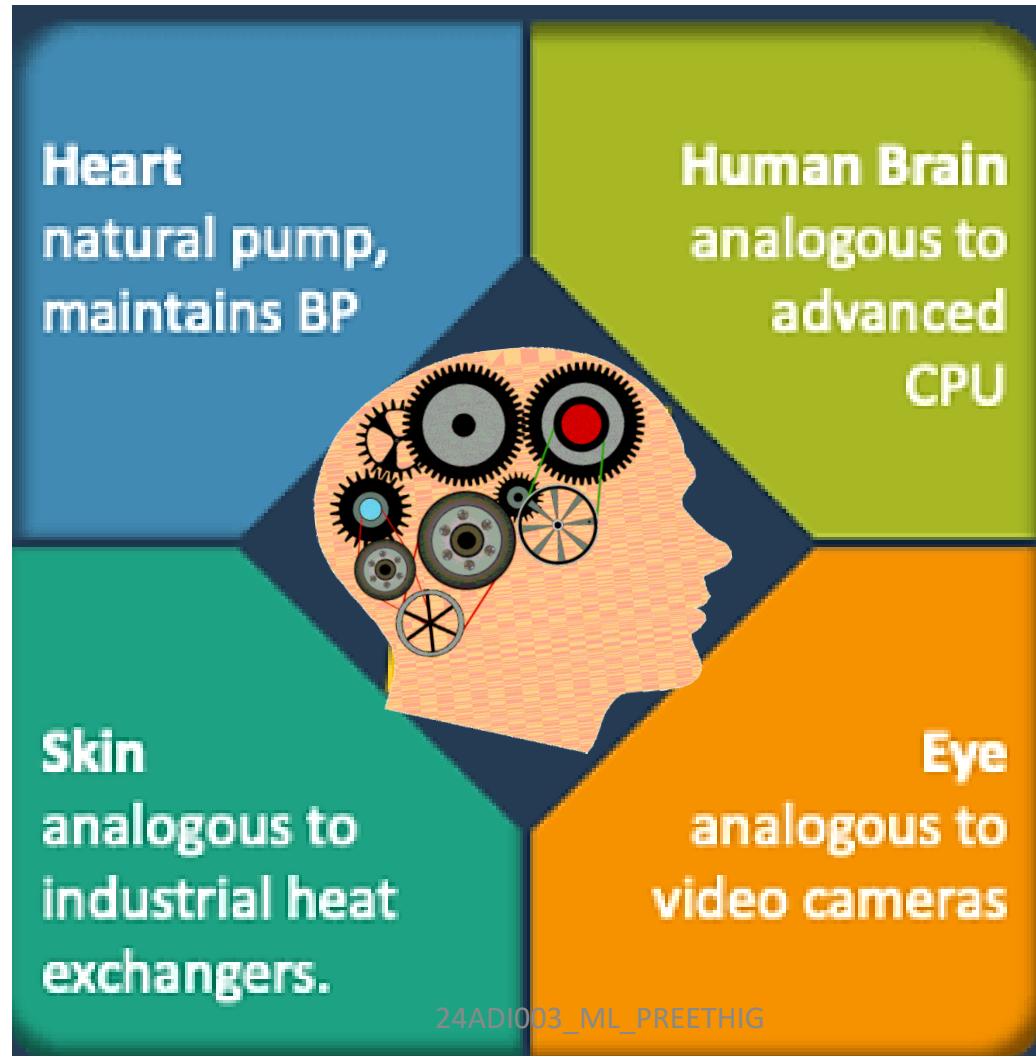
- ✓ ML is not just a tool — it's a **transformational competitive advantage**.
- ✓ Industries that master ML outperform peers in efficiency, insight, and innovation.
- ✓ Real companies are now publicly shifting strategy to be *machine-learning first*.
- ✓ ML expertise is a *core economic driver*, just like oil was historically.

Machine Learning Activity

[Activity Link - Click Here](#)



WHY ARTIFICIAL INTELLIGENCE? ARE HUMANS NOT NATURALLY INTELLIGENT?



Human is a Unique Machine!!!



Human



Computer

Strengths

- Powerful pattern recognition
- Powerful selective attention
- Capacity to learn
- Infinite-capacity LTM
- Rich, multikeyed LTM

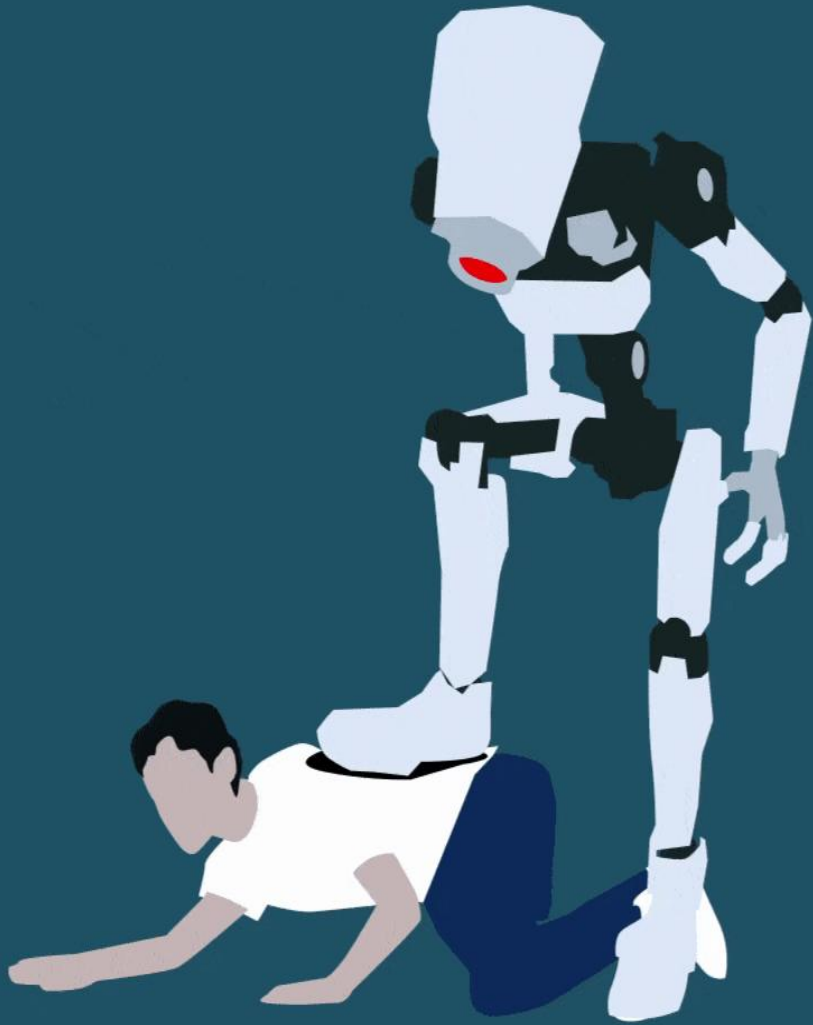
- High-capacity memory
- “permanent” memory
- Very fast processing
- Error-free processing
- Reliable memory access

Weaknesses

- Low-capacity working memory
- Fast-capacity working memory
- Fast-decaying working memory
- Slow-processing
- Error prone processing
- Unreliable success to LTM

- Simple template matching
- Limited learning capacity
- Limited-capacity LTM
- Limited Data integration

Gerd



TEAM ROBOT

24ADI003_ML_PREETHIG

gifs.com



ARTIFICIAL INTELLIGENCE VS HUMAN INTELLIGENCE

	 Human Intelligence	 Artificial Intelligence
 Energy Efficiency	25 watts human brain	2 watts for modern machine learning machine.
 Universal	Humans usually learn how to manage hundreds of different skills during life.	While consuming kilowatts of energy, this machine is usually designed for a few tasks.
 Multi Tasking	Human worker work on multiple responsibilities.	The time needed to teach system on each and every responsibility is considerably high.
 Decision Making	Humans have the ability to learn decision making from experienced scenarios.	Even the most advanced robots can hardly compete in mobility with 6 years old child
 State	Brains are analogue	Computers are digital

AI is needed **ALONGSIDE** Humans
to augment capabilities, **NOT**
REPLACE THEM.

HOW?

PROCESS LARGE DATA



IMPROVE PRECISION



AUTOMATE TASKS



INCREASE EFFICIENCY



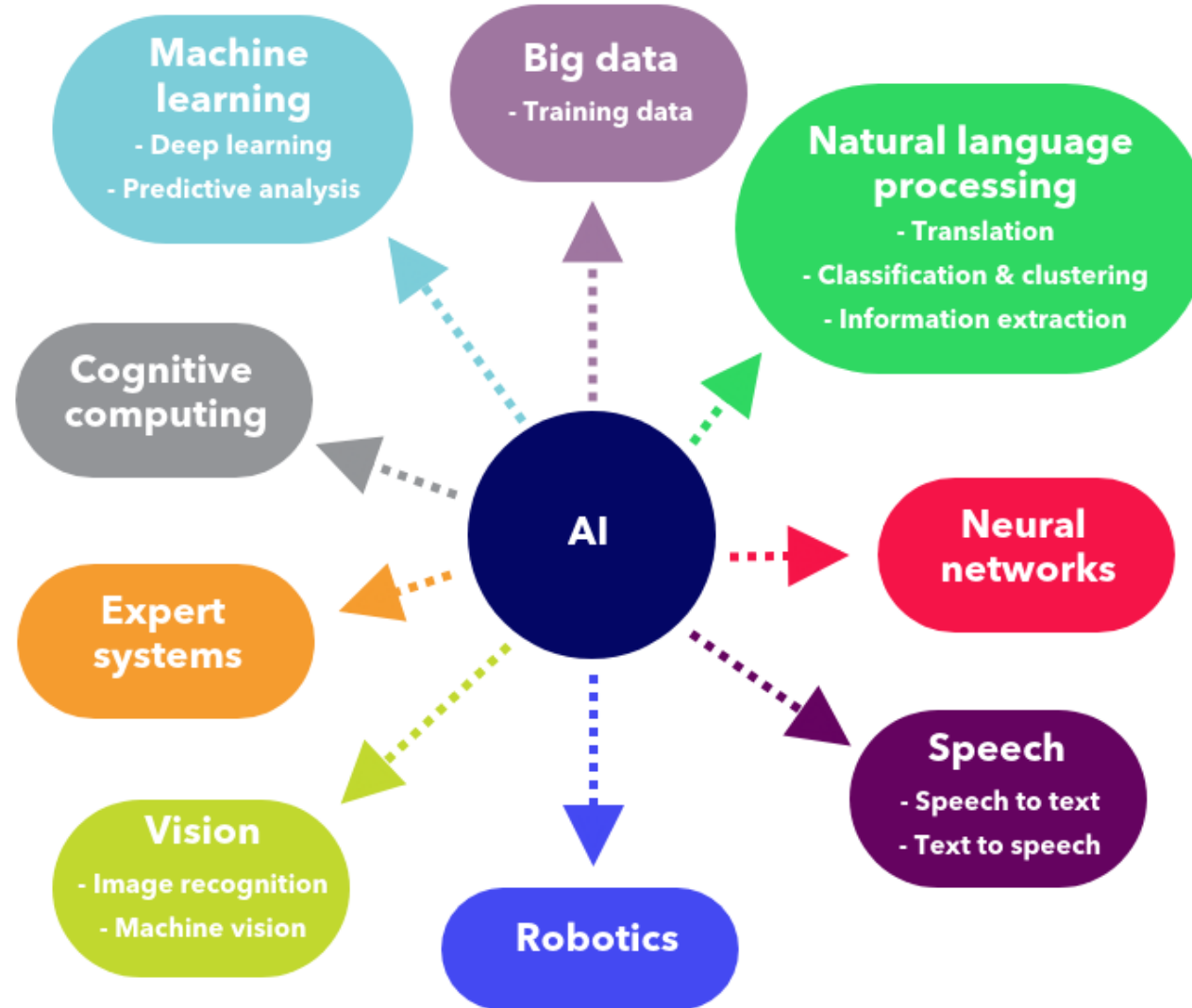
HELPS IN DECISION MAKING



IT'S A WIN-WIN !!!!



THE BRANCHES OF ARTIFICIAL INTELLIGENCE



ARTIFICIAL INTELLIGENCE is any system that achieves one/all of these...

Thinking Humanly

Thinking Rationally

Acting Humanly

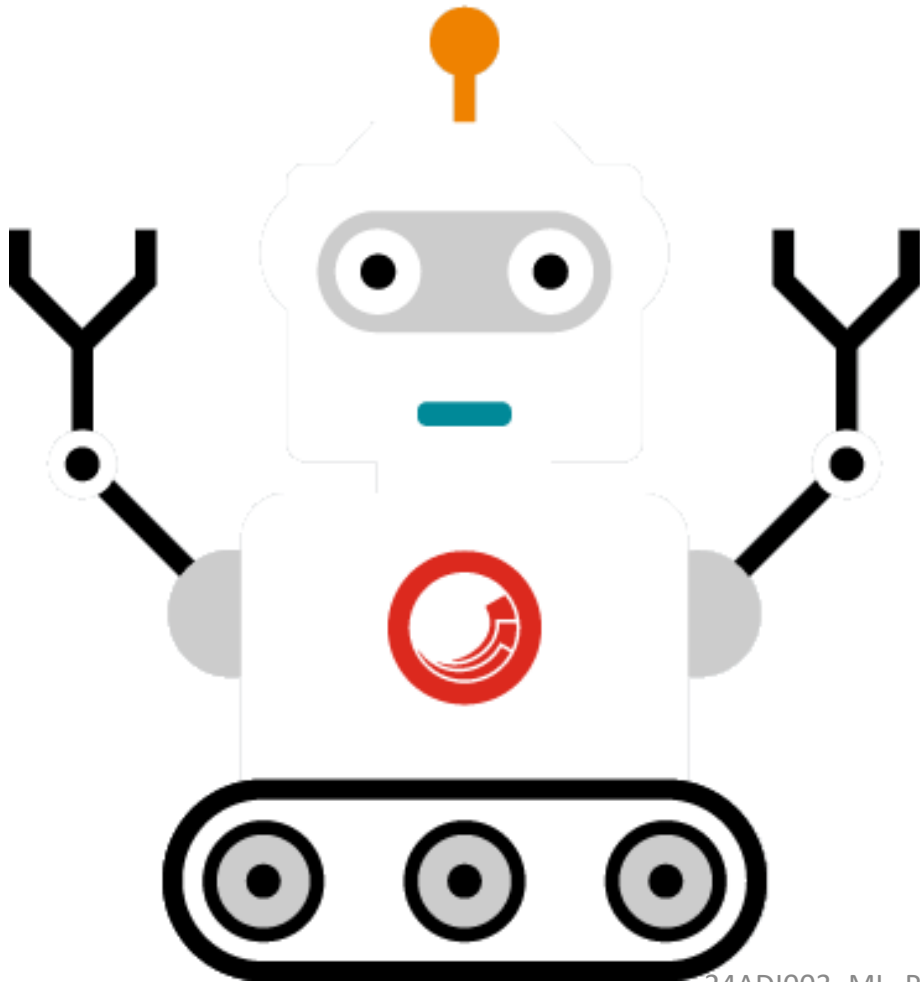
Acting Rationally



How can a Machine do that?



MACHINES LEARN SO THAT THEY CAN BE INTELLIGENT !!!!
JUST THE WAY HUMANS DO!!!!



Remember your FIRST TIME LEARNING ANYTHING?

Recall how you did it.

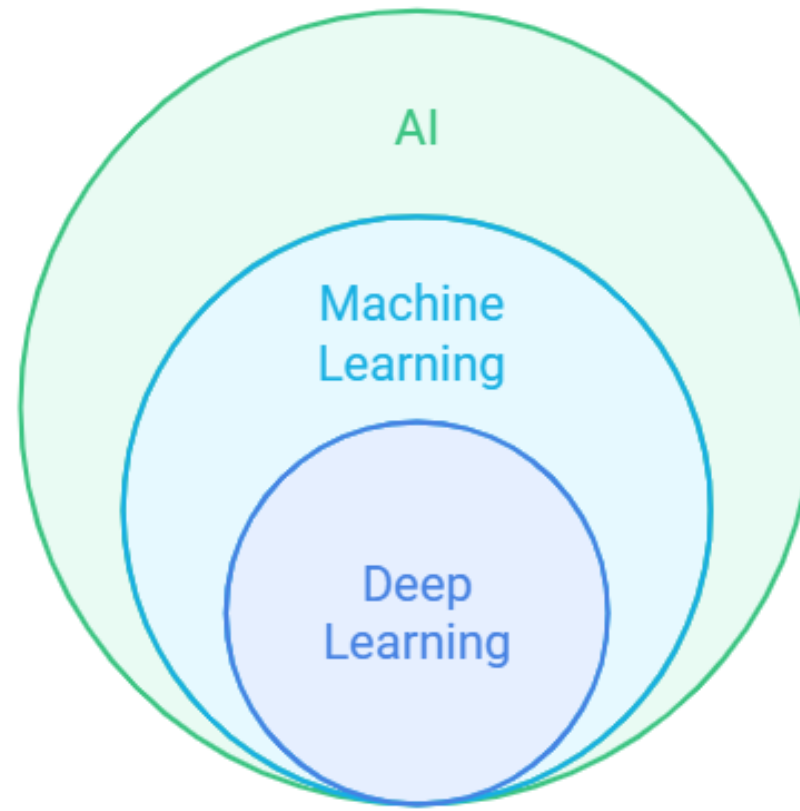
Introduction to Machine Learning

Hierarchy of Learning Technologies

Machines making
decisions

Learns from data to
make decisions

Uses neural networks
for learning



Introduction to Machine Learning

- The term Machine Learning was first coined by **Arthur Samuel** in the year **1959**.
- Machine learning is a branch of artificial intelligence that **enables algorithms to uncover hidden patterns within datasets**.
- It allows them **to predict new, similar data without explicit programming for each task**.
- Machine learning finds **applications** in diverse fields such as image and speech recognition, natural language processing, recommendation systems, fraud detection, portfolio optimization, and automating tasks.

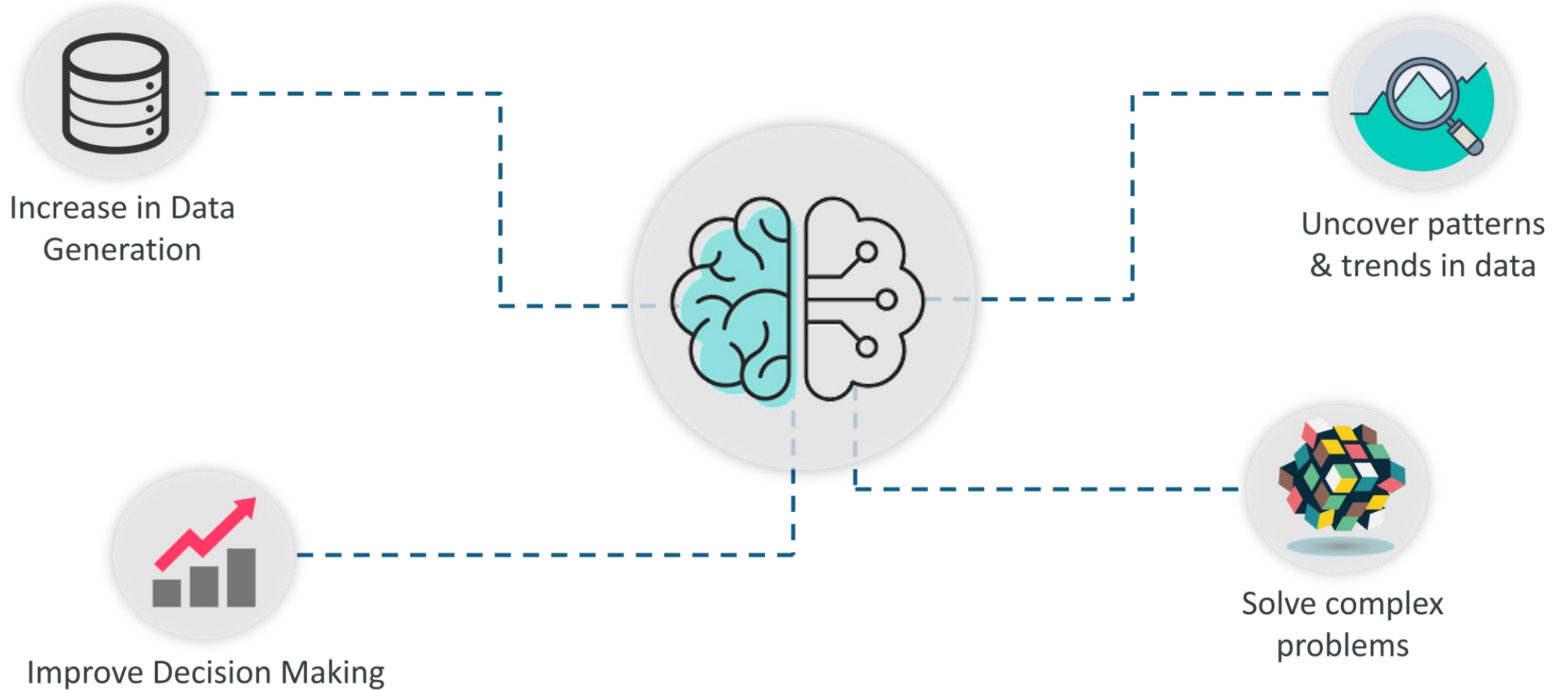
Definitions – Machine Learning

The very first **formal definition** was given by **Tom M. Mitchell**:

“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E.”

In simple terms, Machine learning is a subset of Artificial Intelligence (AI) which provides machines the ability to learn automatically & improve from experience without being explicitly programmed to do so. In the sense, it is the practice of getting Machines to solve problems by gaining the ability to think.

Need for Machine Learning



Need for Machine Learning

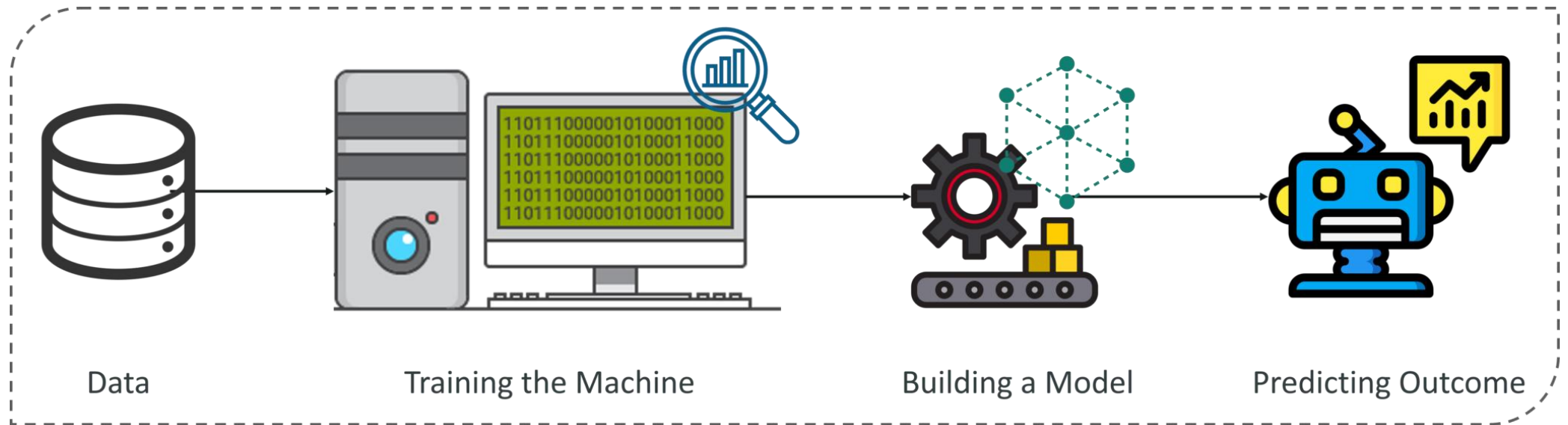
- **Increase in Data Generation:** Due to excessive production of data, we need a method that can be used to structure, analyze and draw useful insights from data. This is where Machine Learning comes in. It uses data to solve problems and find solutions to the most complex tasks faced by organizations.
- **Improve Decision Making:** By making use of various algorithms, Machine Learning can be used to make better business decisions. For example, Machine Learning is used to forecast sales, predict downfalls in the stock market, identify risks and anomalies, etc.

Need for Machine Learning

- **Uncover patterns & trends in data:** Finding hidden patterns and extracting key insights from data is the most essential part of Machine Learning. By building predictive models and using statistical techniques, Machine Learning allows you to dig beneath the surface and explore the data at a minute scale. Understanding data and extracting patterns manually will take days, whereas Machine Learning algorithms can perform such computations in less than a second.
- **Solve complex problems:** From detecting the genes linked to the deadly ALS disease to building self-driving cars, Machine Learning can be used to solve the most complex problems.

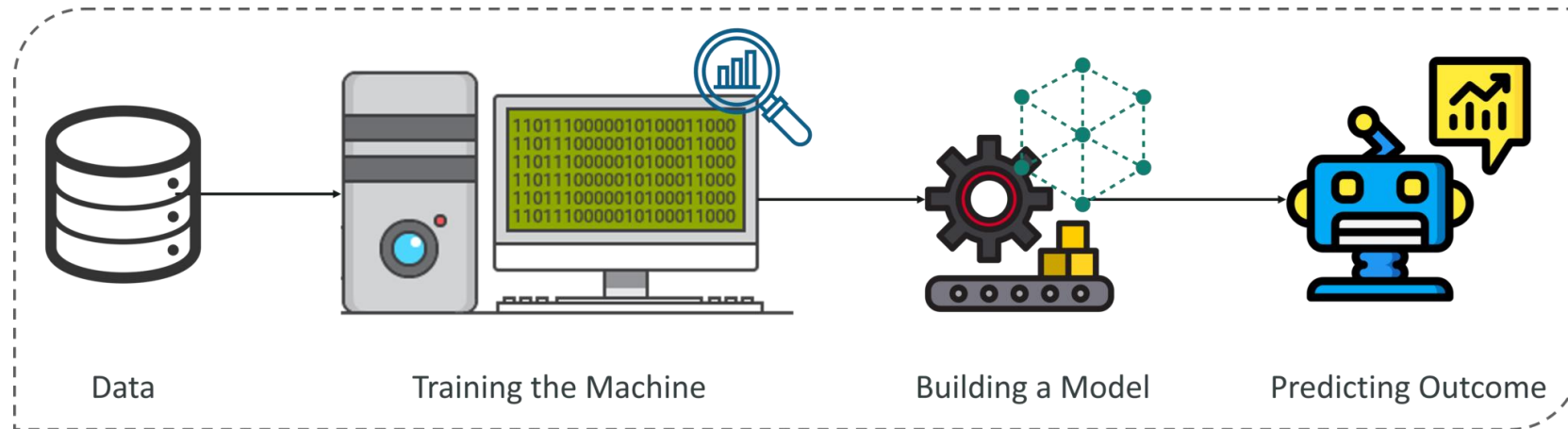
Machine Learning Process

- A Machine Learning process **begins by feeding the machine lots of data**; by using this data the machine is trained to detect hidden insights and trends.
- These insights are then used to build a **Machine Learning Model** by using an algorithm in order to solve a problem.



Machine Learning Process

- A **machine "learns"** by identifying patterns in data and improving its **ability to perform specific tasks** without being explicitly programmed for every scenario.
- This learning process **helps machines to make accurate predictions or decisions** based on the information they receive.
- Unlike traditional programming where instructions are fixed, **ML allows models to adapt and improve through experience.**



Machine Learning Process

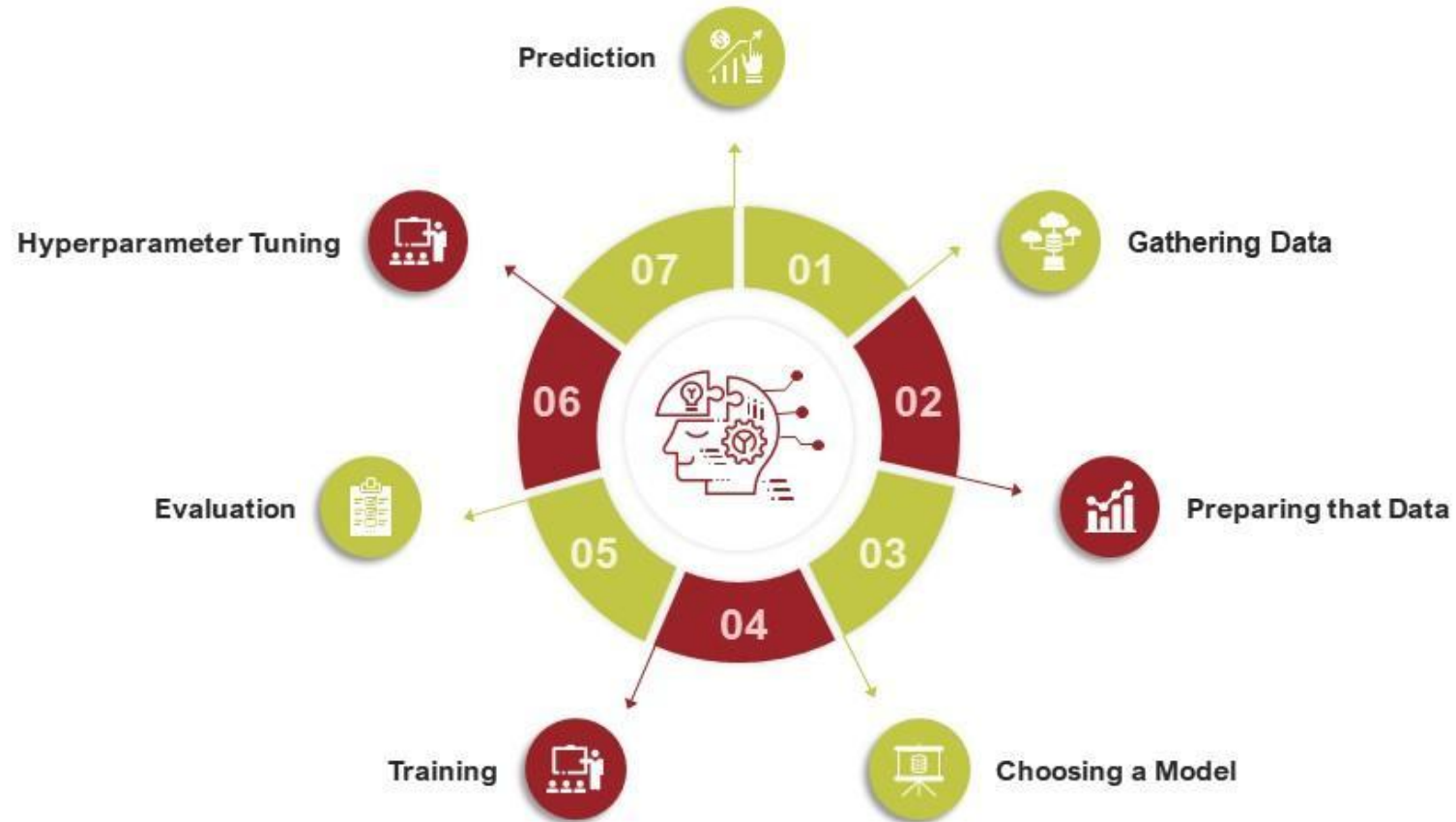
Here is how the learning process works:

1. **Data Input:** Machine needs data like text, images or numbers to analyze. Good quality and enough quantity of data are important for effective learning.
2. **Algorithms:** Algorithms are mathematical methods that help the machine find patterns in data. Different algorithms help different tasks such as classification or regression.
3. **Model Training:** During training, the machine adjusts its internal settings to better predict outcomes. It learns by reducing the difference between its predictions and actual results.

Machine Learning Process

4. **Feedback Loop:** Machine compares its predictions with true outcomes and uses this feedback to correct errors. Techniques like gradient descent help it update and improve.
5. **Experience and Iteration:** Machine repeats training many times with data helps in refining its predictions with each pass, more data and iterations improve accuracy.
6. **Evaluation and Generalization:** Model is tested on unseen data to ensure it performs well on real-world tasks.

7 Steps of Machine Learning



Importance of Data in Machine Learning

Data is the foundation of machine learning (ML) without quality data ML models cannot learn, perform or make accurate predictions.

- Data provides the examples from which models learn patterns and relationships.
- High-quality and diverse data improves how well models perform and generalize to new situations.
- It helps models to understand real-world scenarios and adapt to practical uses.
- Features extracted from data are important for effective training.
- Separate datasets for validation and testing measure how well the model works on unseen data.
- Data drives continuous improvements in models through feedback loops.



Machine Learning Terms & Definitions

Machine Learning - Terms & Definitions

- **Algorithm:** A Machine Learning algorithm is a set of rules and statistical techniques used to learn patterns from data and draw significant information from it. It is the logic behind a Machine Learning model. An example of a Machine Learning algorithm is the Linear Regression algorithm.
- **Model:** A model is the main component of Machine Learning. A model is trained by using a Machine Learning Algorithm. An algorithm maps all the decisions that a model is supposed to take based on the given input, in order to get the correct output.
- **Predictor Variable:** It is a feature(s) of the data that can be used to predict the output.

Machine Learning - Terms & Definitions

- **Response Variable:** It is the feature or the output variable that needs to be predicted by using the predictor variable(s).
- **Training Data:** The Machine Learning model is built using the training data. The training data helps the model to identify key trends and patterns essential to predict the output.
- **Testing Data:** After the model is trained, it must be tested to evaluate how accurately it can predict an outcome. This is done by the testing data set.

LEARNING
IS
THE
KEY
TO
SUCCESS

Humans don't have only one type of learning everything.

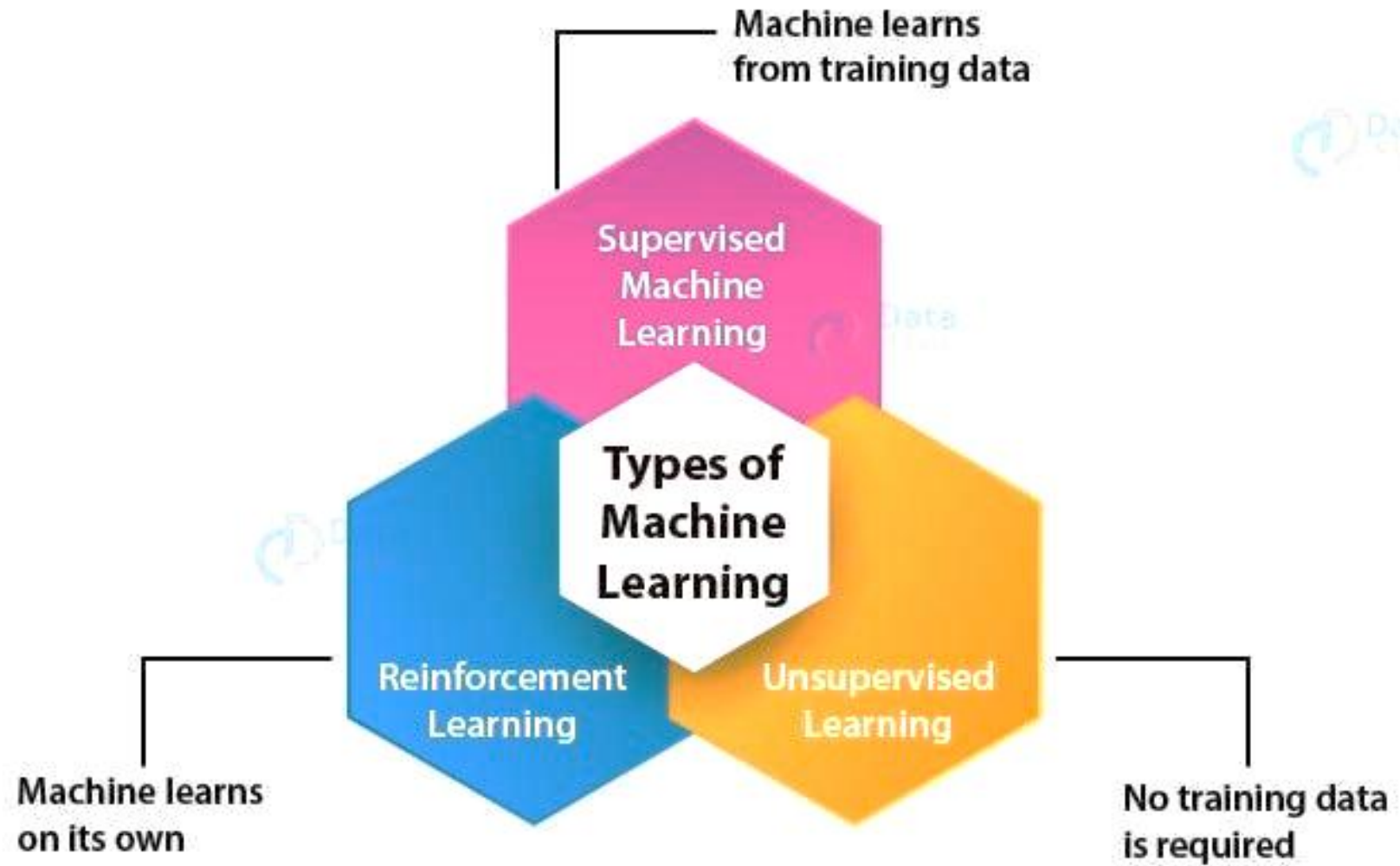
So Why should machines restrict to one type of learning

Types of Machine Learning

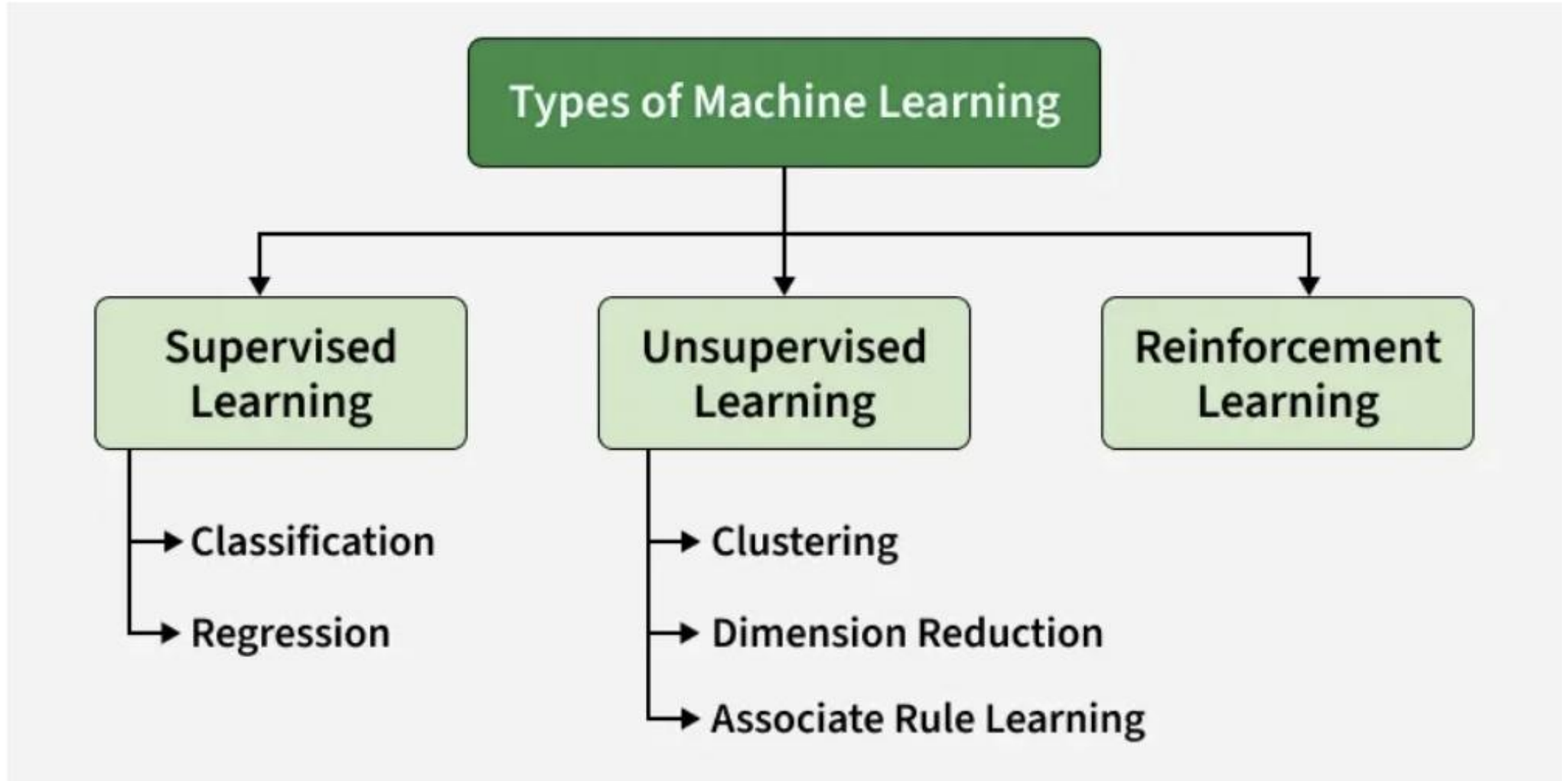
There are several types of machine learning, each with special characteristics and applications. Some of the main types of machine learning algorithms are as follows:

- Supervised Machine Learning
- Unsupervised Machine Learning
- Reinforcement Learning

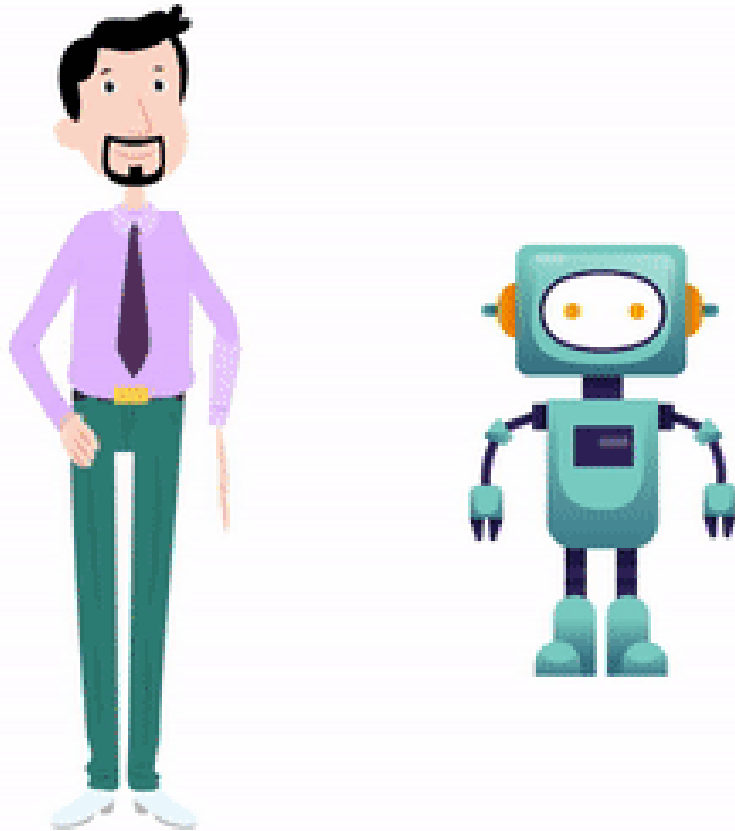
Additionally, there is a more specific category called **Semi-Supervised Learning and Self-Supervised Learning**, which combines elements of both supervised and unsupervised learning



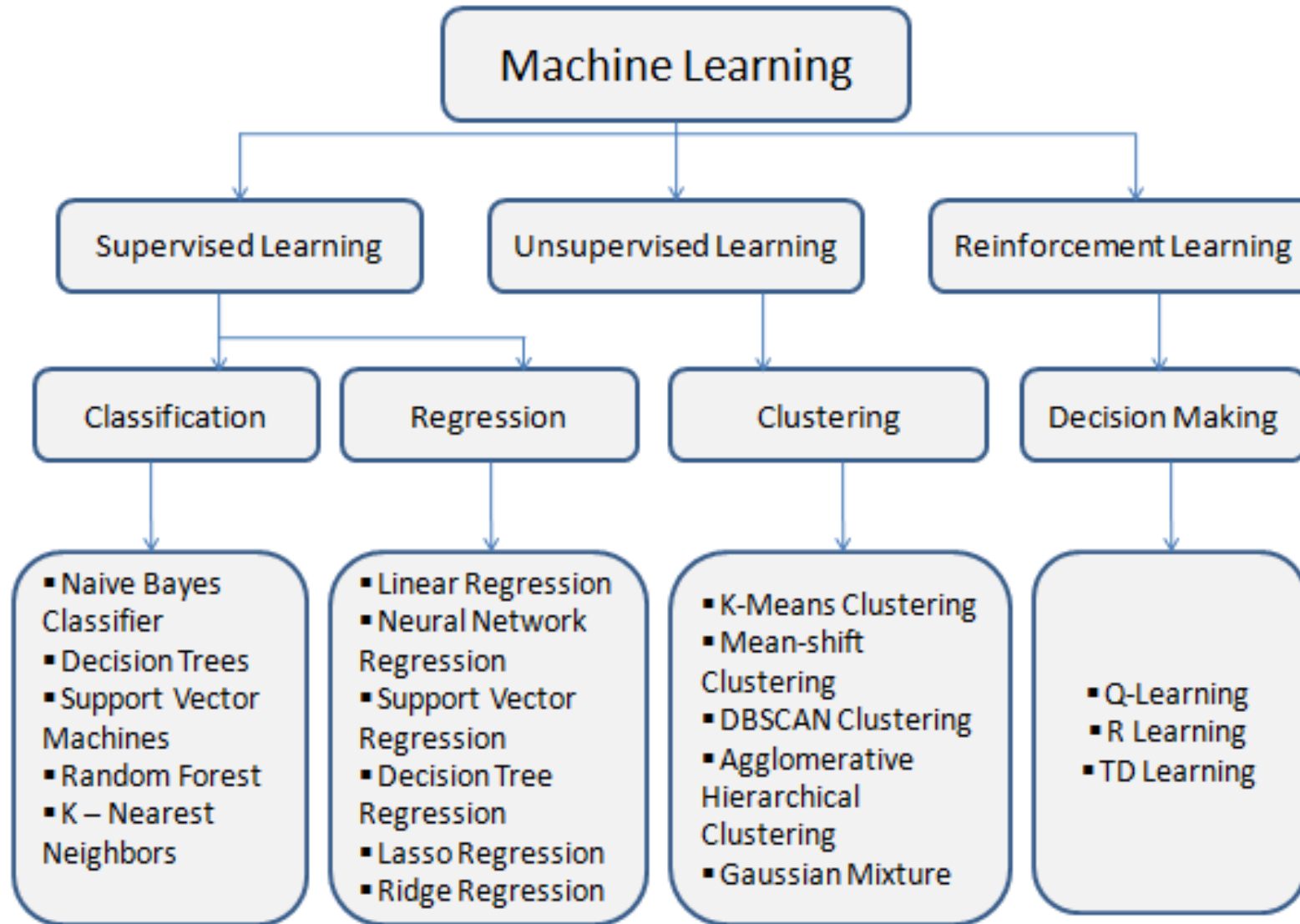
Types of Machine Learning



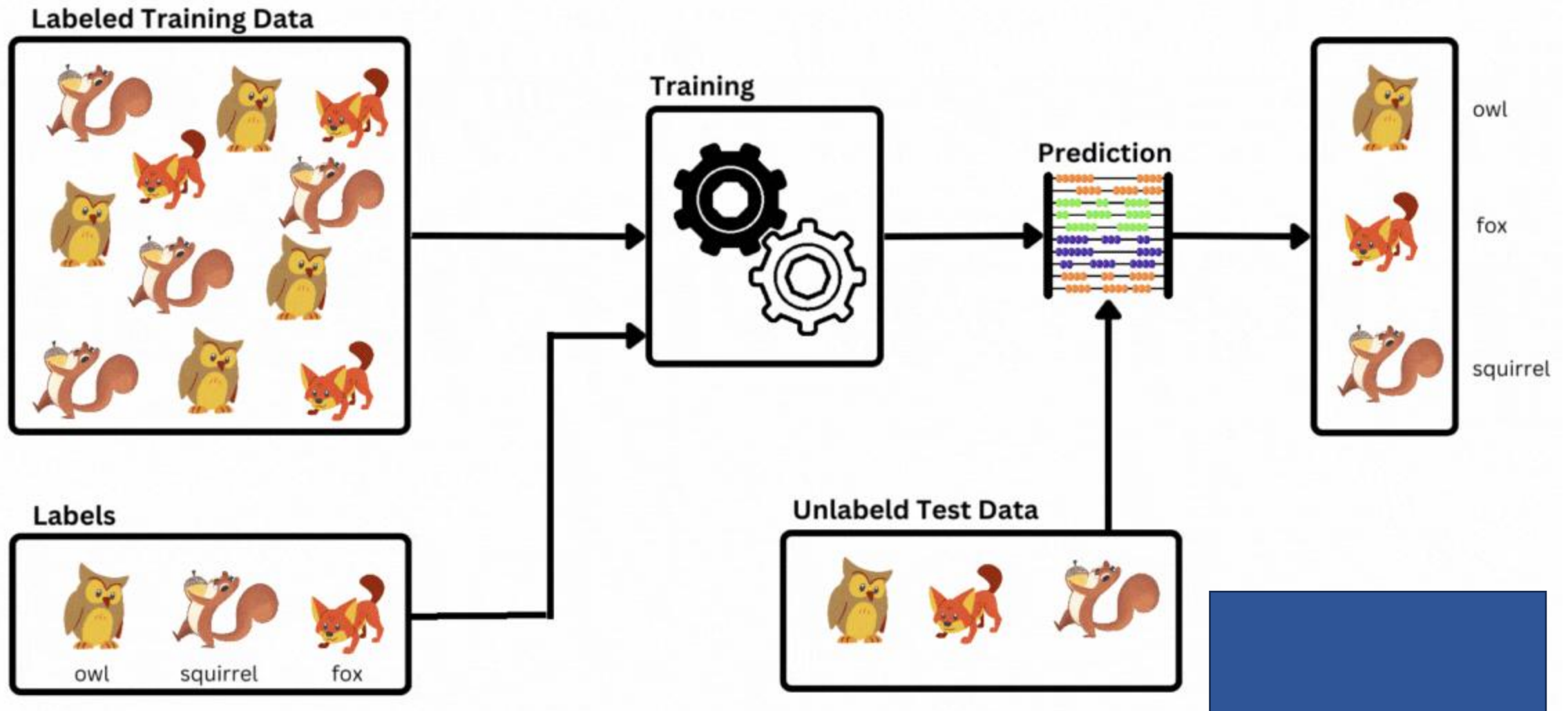
Supervised Learning



Types of Machine Learning



Supervised Learning



Supervised Machine Learning

- **Supervised learning** is defined as when a **model gets trained on a "Labeled Dataset"**.
- Labelled datasets have both input and output parameters.
- Supervised Learning algorithms learn to map points between inputs and correct outputs.
- It has **both training and validation datasets labelled**.

Supervised Machine Learning

- **Example:** If you train a model using labeled images of cats and dogs, it learns the features of each. When shown a new image, it predicts whether it's a cat or a dog.
- There are two main categories of supervised learning that are mentioned below:
 - ✓ Classification
 - ✓ Regression

Supervised Machine Learning

Classification predicts categorical outputs, meaning it assigns data into predefined classes like spam/non-spam emails or disease risk categories. These algorithms learn to map input features to discrete labels. Here are some classification algorithms:

- Logistic Regression
- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Naive Bayes
- Support Vector Machine

Supervised Machine Learning

Regression predicts continuous values, such as house prices or product sales. It learns the relationship between input features and a numerical target variable. Here are some regression algorithms:

- Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression
- Decision tree
- Random Forest

Where to Use Supervised Learning?

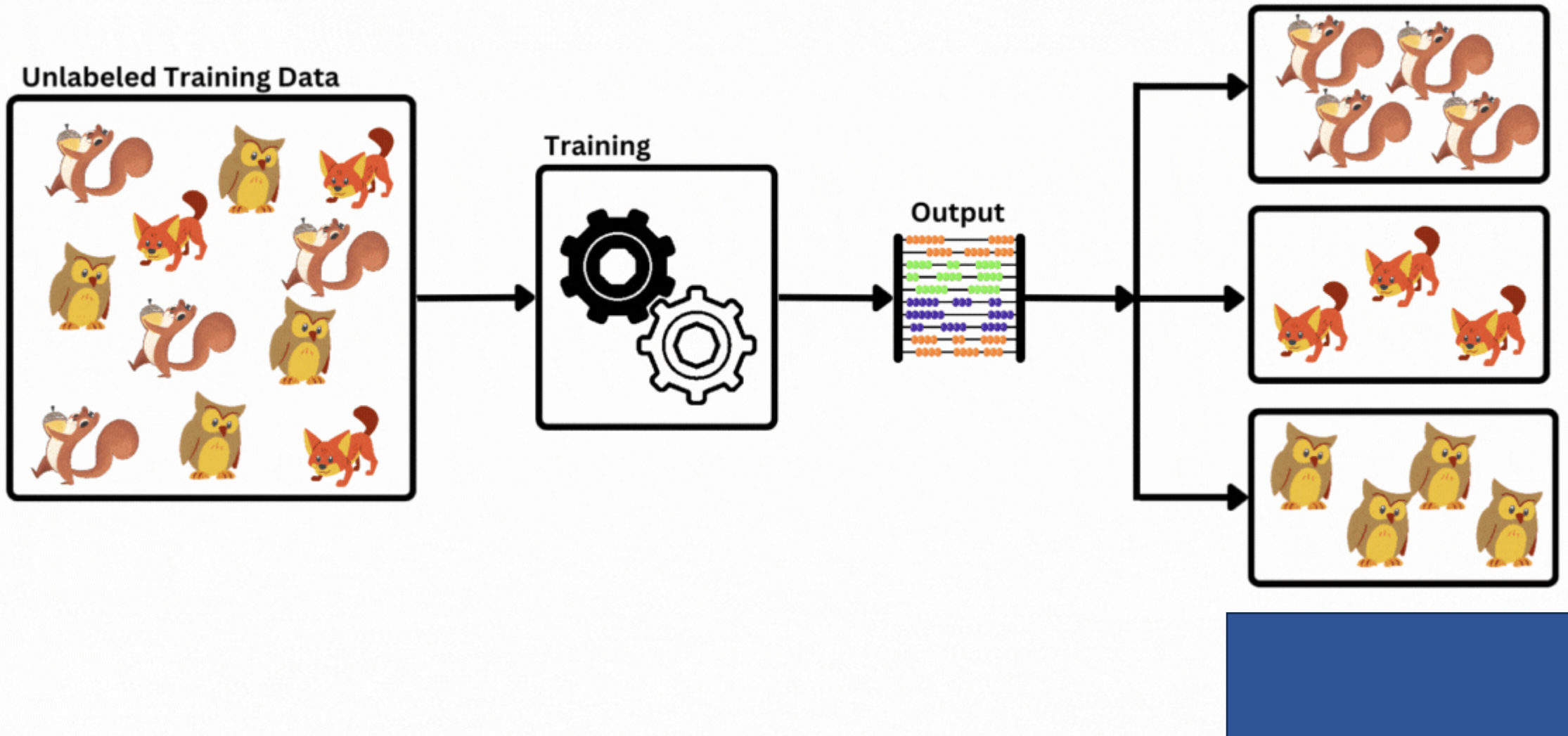
- When you have labeled data and want to predict outcomes.
- Ideal for classification (like spam detection) or regression tasks (like price forecasting).
- Best used in domains where historical data with outcomes is already available.

Applications of Supervised Learning

Supervised learning is used in a wide variety of applications, including:

- **Image, speech and text processing:** For tasks like image classification, speech recognition and sentiment analysis.
- **Predictive analytics:** To forecast sales, customer churn, stock prices and weather conditions.
- **Recommendation and personalization:** Powering systems that suggest products, movies or content.
- **Healthcare and finance:** Used for medical diagnosis, fraud detection and credit scoring.
- **Automation and control:** In autonomous vehicles, manufacturing quality checks and gaming AI.

Unsupervised Learning



Unsupervised Machine Learning

- **Unsupervised Learning** works with **unlabeled data**, meaning there are no **predefined outputs**.
- The algorithm finds hidden patterns, groups or relationships within the data on its own.
- It's mainly used for **clustering, dimensionality reduction and data visualization**.

Unsupervised Machine Learning

- **Example:** If you have customer data without labels, the algorithm can group similar customers based on purchase behavior useful for segmentation and marketing.
- There are two main categories of unsupervised learning that are mentioned below:
 - ✓ Clustering
 - ✓ Dimensionality Reduction Techniques
 - ✓ Association Rule Learning

Unsupervised Machine Learning

- **Clustering** is the process of **grouping data points into clusters based on their similarity.**
- This technique is useful for **identifying patterns and relationships** in data without the need for labeled examples.
- Common techniques include:
 - ✓ K-Means
 - ✓ DBSCAN
 - ✓ Mean-shift

Unsupervised Machine Learning

- **Dimensionality reduction** helps reduce the number of features while preserving important information.
- Common techniques include:
 - ✓ Principal Component Analysis
 - ✓ Independent Component Analysis

Unsupervised Machine Learning

- **Association rule learning** is a technique for **discovering relationships between items in a dataset.**
- It identifies rules that indicate the presence of one item implies the presence of another item with a specific probability.
- Common techniques include:
 - ✓ Apriori
 - ✓ FP-growth
 - ✓ Eclat

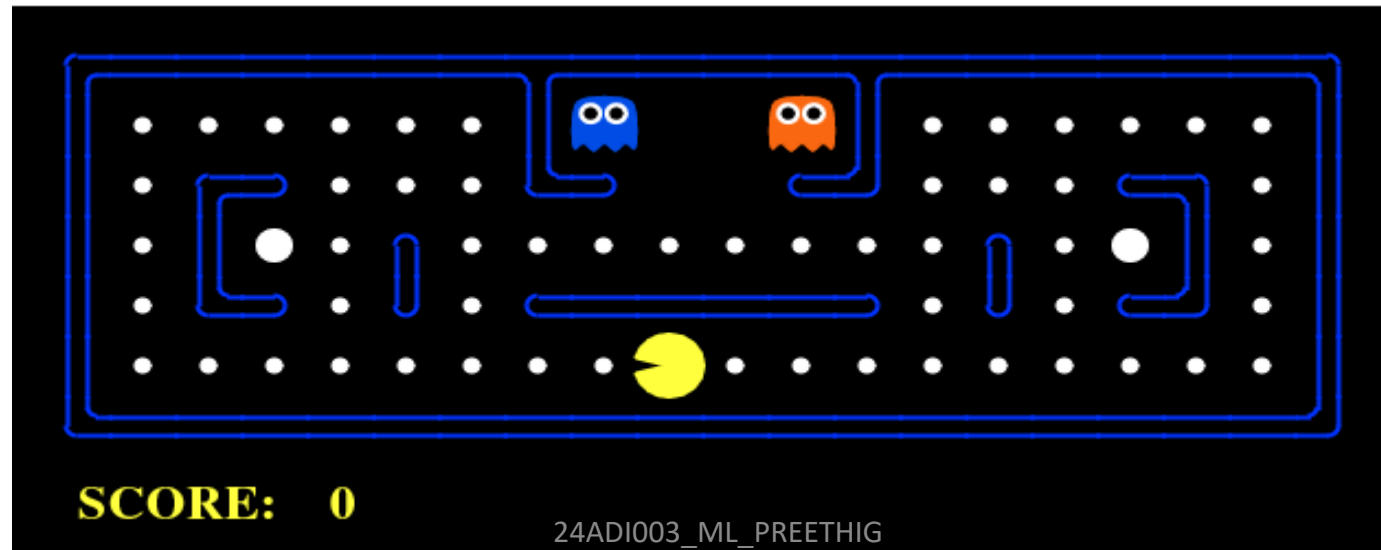
Where to Use Unsupervised Learning?

- When data is unlabeled or unstructured.
- Useful for exploratory analysis, clustering or feature extraction.
- Common in marketing, recommendation systems and fraud detection where patterns matter more than labels.

Applications of Unsupervised Learning

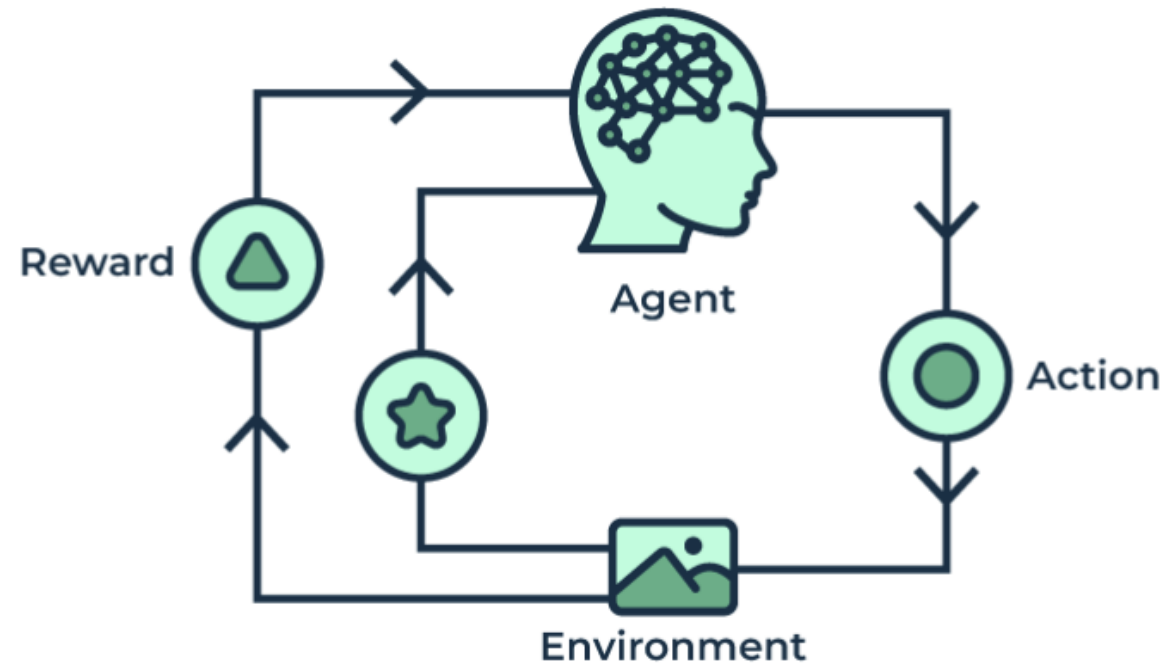
- **Clustering and segmentation:** Group similar data points, customers or images.
- **Anomaly detection:** Spot unusual patterns or outliers in data.
- **Dimensionality reduction:** Simplify large datasets while retaining key information.
- **Recommendation and marketing:** Identify user preferences and improve product suggestions.
- **Data preprocessing and analysis:** Clean data, detect patterns and support exploratory data analysis (EDA).

Reinforcement Learning



Reinforcement Learning

- Reinforcement learning **trains an agent** to make a sequence of decisions through trial and error.
- The agent interacts with the environment, receives feedback in the form of rewards or penalties and learns optimal actions over time.



Reinforcement Learning

- **Example:** An AI agent learning to play chess gets positive feedback for good moves and negative for poor ones. Over time, it learns strategies to win more often.
- Here are some of most common reinforcement learning algorithms:
 - ✓ **Q-learning:** Learns the best action for each state based on expected rewards.
 - ✓ **SARSA (State-Action-Reward-State-Action):** Similar to Q-learning but updates values for the action actually taken.
 - ✓ **Deep Q-learning:** Uses neural networks to handle complex state-action relationships.

Types of Reinforcement Learning

- **Positive Reinforcement:** Rewards desired behavior (e.g., giving points for correct answers).
- **Negative Reinforcement:** Removes negative outcomes to encourage good actions (e.g., turning off a buzzer after the right move).

Where to Use Reinforcement Learning?

- When you need an agent to learn by interacting with an environment.
- Best for decision-making or optimization tasks involving trial and feedback loops.
- Used when long-term performance or adaptive behavior is more important than immediate accuracy.

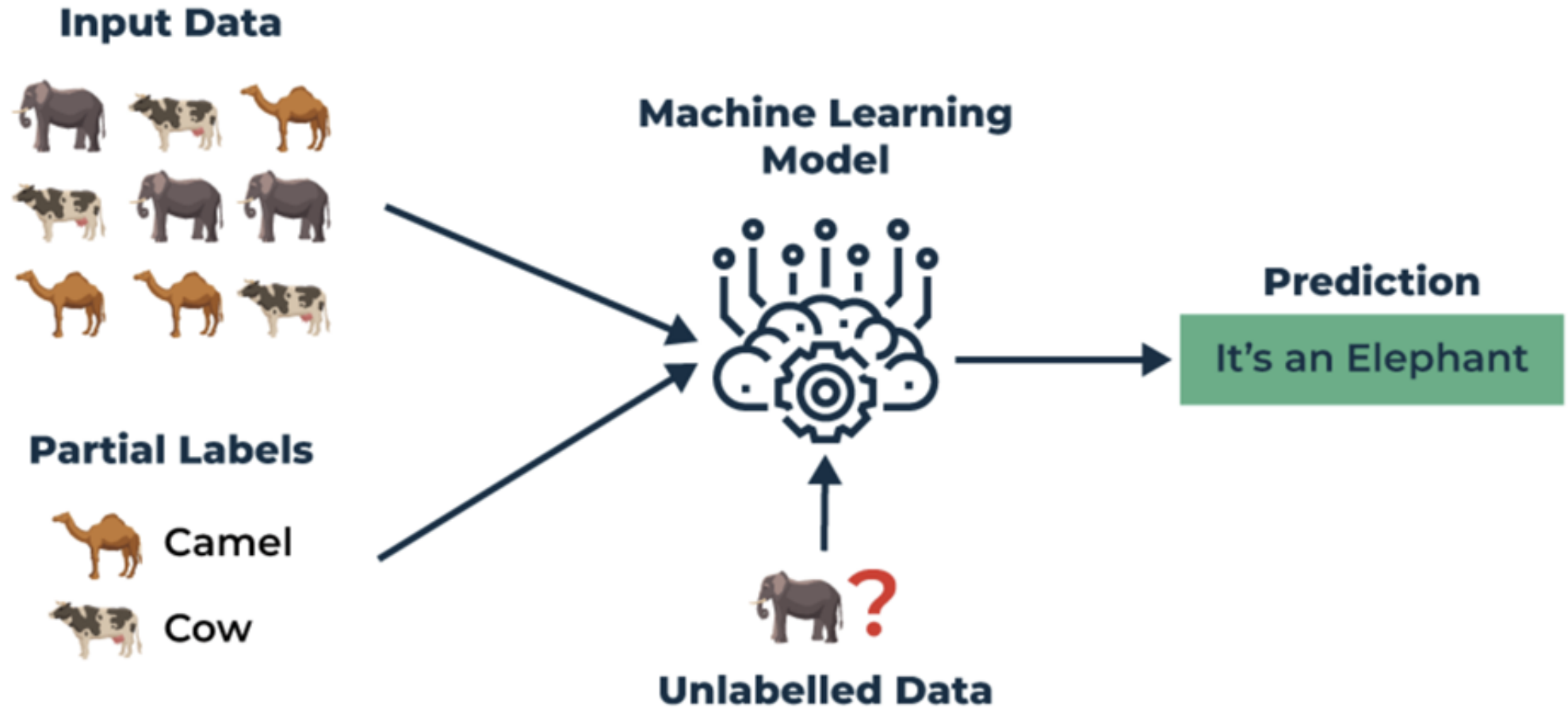
Applications of Reinforcement Learning

- **Gaming and simulation:** Teaching agents or NPCs to play and adapt intelligently.
- **Robotics and automation:** Enabling robots to perform tasks autonomously.
- **Autonomous vehicles:** Helping self-driving cars make real-time decisions.
- **Healthcare and finance:** Optimizing treatment plans, trading and resource allocation.
- **Recommendation and personalization:** Improving user experience through adaptive suggestions.
- **Industrial and energy management:** Optimizing control systems and energy use.

Semi-Supervised Learning

- Semi-Supervised learning Semi-Supervised Learning **combines both Supervised and Unsupervised approaches.**
- It uses a **small set of labeled data and a large set of unlabeled data** for training useful when labeling is costly or time-consuming.
- **Example:** Consider that we are building a language translation model, having labeled translations for every sentence pair can be resources intensive. **It allows the models to learn from labeled and unlabeled sentence pairs, making them more accurate.** This technique has led to significant improvements in the quality of machine translation services.

Semi-Supervised Learning



Semi-Supervised Learning

Popular Techniques

- **Graph-based Learning:** Spreads label information through data relationships.
- **Label Propagation:** Iteratively assigns labels to unlabeled data.
- **Co-training:** Uses two models to train and label each other's data.
- **Self-training:** Uses model predictions as pseudo-labels.
- **Generative Adversarial Networks (GANs):** Generates synthetic data to improve learning.

Where to Use Semi-Supervised Learning?

- When you have limited labeled data but plenty of unlabeled data.
- Useful for domains with high labeling costs, such as medical, NLP or image datasets.
- Ideal when unlabeled data still holds valuable information that can improve learning performance.

Applications of Semi-Supervised Learning

- **Image Classification:** Combine small labeled and large unlabeled image datasets to improve accuracy.
- **Natural Language Processing (NLP):** Enhance language models by using a mix of labeled and vast unlabeled text data.
- **Speech Recognition:** Boost accuracy by leveraging limited transcribed audio and more unlabeled speech data.
- **Recommendation Systems:** Improve recommendations using sparse labeled data and abundant unlabeled user behavior.
- **Healthcare & Medical Imaging:** Improve medical image analysis with a mix of labeled and unlabeled images.

Self-Supervised Learning

- Self-Supervised Learning (SSL) is a modern approach where models generate their own labels from raw data. It doesn't rely on manual annotation instead, the model learns by predicting parts of data from other parts.
- **Example:** In NLP, models like BERT or GPT learn by predicting masked words in sentences, using surrounding context as supervision. This helps them learn language understanding without human labeling.

Self-Supervised Learning

Popular Techniques

- Masked Modeling (BERT)
- Contrastive Learning (SimCLR, MoCo)
- Autoencoders
- Predictive Coding

Applications

- Natural Language Processing
- Computer Vision and Speech Recognition
- Video understanding
- Pre-training for large AI models

Where to Use Self-Supervised Learning?

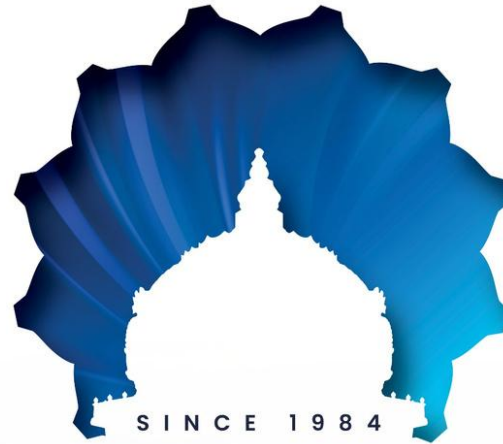
- When manual labeling is impossible or expensive.
- Suitable for large-scale datasets like text, audio and images.
- Best for pre-training models that can later be fine-tuned for specific supervised tasks.

ML Types - Comparison

Type	Data Requirement	Label Availability	Learning Goal	Common Use Case
Supervised	High	Labeled	Predict outputs	Spam detection, Price prediction
Unsupervised	Medium	Unlabeled	Find hidden patterns	Customer segmentation
Reinforcement	High	Reward feedback	Learn best actions	Robotics, Games
Semi-Supervised	Medium	Partial labels	Combine both learning types	NLP, Image recognition
Self-Supervised	High	Self-generated labels	Learn data representations	BERT, GPT, CLIP

References

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- <https://www.geeksforgeeks.org/machine-learning/introduction-machine-learning/>
- <https://www.geeksforgeeks.org/machine-learning/types-of-machine-learning/>
- <https://www.analyticsvidhya.com/blog/2021/03/everything-you-need-to-know-about-machine-learning/>



FOUR DECADES OF NATION BUILDING THROUGH EXCELLENCE IN EDUCATION

KUMARAGURU

ENGINEERING | LIBERAL ARTS | AGRICULTURE | MANAGEMENT
BUSINESS INCUBATION | TAMIL RESEARCH



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COLLEGE OF TECHNOLOGY
character is life



KCT
BUSINESS SCHOOL



KUMARAGURU
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ARTS AND SCIENCE



KUMARAGURU
INSTITUTE OF
AGRICULTURE



KUMARAGURU
SCHOOL OF BUSINESS



KRIA
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RESEARCH & INNOVATION
ALLIANCE



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INNOVATION



KUMARAGURU | **KCI**
CENTRE FOR INDUSTRIAL
RESEARCH & INNOVATION

FORGE 



SEA | SAKTHI
EXCELLENCE
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SAKTHI INTERNATIONAL EDUCATION



N.MAHALINGAM
TAMIL RESEARCH
CENTRE



KCT | TECH PARK



KARE
Kumaraguru
Action For
Relief And Empowerment



Dr.N.MAHALINGAM
CHESS ACADEMY

THANK YOU
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